

Morphological and genomic characterization of a novel *Hepatozoon* lineage from the critically endangered insular pitviper *Bothrops sazimai* (Squamata: Serpentes: Viperidae)

Genetica

71°CBG

2026

International Congress of the Brazilian Genetics Society

CentroSul Florianópolis - SC September 29 to October 2, 2026

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In the service of life

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Introduction

Hepatozoon (Apicomplexa: Adeleorina: Hepatozoidae) is the blood parasite most often reported in snakes. Most species were described from gamont shape and short 18S rRNA fragments. Recent studies in South America keep finding lineages that morphology alone cannot separate.

The Laboratório de Ecologia e Evolução (Instituto Butantan) keeps critically endangered snakes for conservation. We found *Hepatozoon* by chance in blood smears of *Bothrops sazimai* prepared for flow cytometry and genome studies. *B. sazimai* lives only on Ilha dos Franceses, Espírito Santo, Brazil (Fig. 1)¹. Rising sea levels cut the island off from the mainland within the last 11,000 to 55,000 years².

Objective. Characterize a *Hepatozoon* lineage found in *B. sazimai* by gamont morphology, 18S rRNA, and complete mitochondrial and apicoplast genomes, and compare it with known species.

Methods

We scanned Giemsa-stained blood smears of an adult female (Zeiss AxioScan Z1), measured gamonts in QuPath, and compared them with 28 records of snake *Hepatozoon* from 14 studies. We pulled parasite reads from the host's Nanopore genome assembly (Fig. 2) and placed the 18S variants in an IQ-TREE maximum likelihood tree of 150 adeleorin sequences.

The diagram illustrates the experimental workflow. The top section, 'gamont morphometry', shows a snake being bled, followed by a blood smear, slide scanning, and morphometric analysis. The bottom section, 'parasite genomics', shows DNA extraction and nanopore sequencing, followed by quality trimming and initial assembly, apicomplexa/hepatozoon screening, read mapping/assembly, and gene annotation. Key tools and databases mentioned include PromethION, Flye 2.9.3, NCBI databases, Minimap2, and Geneious.

Figure 2. Workflow: gamont morphometry (top) and parasite genomics (bottom). Created in BioRender (BioRender.com).

Results

Short gamonts with a long, narrow nucleus

Gamonts measured $11.9 \pm 1.3 \mu\text{m}$ by $4.3 \pm 0.7 \mu\text{m}$, with a nucleus of $5.0 \pm 0.9 \mu\text{m}$ by $2.4 \pm 0.6 \mu\text{m}$ (Fig. 3).

A micrograph showing Giemsa-stained blood smears. Two erythrocytes are visible, each carrying a curved Hepatozoon gamont. The gamonts are positioned next to the nuclei of the erythrocytes.

Figure 3. Giemsa-stained blood smear of *B. sazimai*. Two erythrocytes carry curved *Hepatozoon* gamonts beside the nucleus.

Only *H. atrocis* has shorter gamonts among the compared species (Fig. 4A). The gamont is short and broad (length/width 2.8), but its nucleus is long and narrow (length/width 2.1). No compared species combines the two (Fig. 4B).

Two 18S variants that do not group together

The host carries two 18S rRNA variants: V1 is dominant (944 reads) and V2 is minor (271 reads). Both fall inside *Hepatozoon* (Fig. 5). V2 sits in a clade of Neotropical snake parasites (SH-aLRT 97.7, UFBoot 100), next to *H. annulatum*. V1 falls outside that clade, as sister to a group of snake and turtle parasites.

Figure 4 consists of two scatter plots. Plot A shows gamont size (length vs. width) for various species. Plot B shows the ratio of nucleus length to width versus gamont length/width. The 'THIS STUDY' data points are highlighted in red. The legend lists species: H. angeladaviesae, H. annulatum, H. atrocis, H. ayorgbor, H. boigae, H. carinacauda, H. cecilioharei, H. cevapii, H. cuetensis, H. massardii, H. musa, H. quagliattus, H. sipedon, Hepatozoon sp., H. trigeminum, H. viperai.

Figure 4. Gamonts of snake *Hepatozoon*. A: size (mean $\pm$ SD). B: gamont and nucleus shape.

A phylogenetic tree showing the relationships between 150 adeleorin 18S rRNA sequences. The tree is rooted on Adelina. Node labels indicate SH-aLRT/ultrafast bootstrap values. Triangles represent collapsed clades. Red highlights the two variants from B. sazimai (V1 and V2). The scale bar represents substitutions per site.

Figure 5. Maximum likelihood tree of 150 adeleorin 18S rRNA sequences, rooted on *Adelina*. Node labels: SH-aLRT/ultrafast bootstrap. Triangles: collapsed clades. Red: V1 and V2, the two variants from *B. sazimai*. Scale bar: substitutions per site.

A large mitochondrial genome with a new gene order

The mitochondrial genome is a circular molecule of 10,479 bp, 63% longer than the largest published congener (6,413 bp). It carries *cox1*, *cox3*, *cob*, and 25 rRNA fragments, like its relatives. The extra length sits between *cox1* and *cob*, where *cox3* now follows *cob* (Fig. 7). The apicoplast genome is 32,316 bp, close to the only other published *Hepatozoon* apicoplast genome (*H. canis*, 31.9 kb). It carries 22 protein-coding genes, 26 tRNA genes, and 4 rRNA genes (Fig. 6).

A linear representation of the apicoplast genome. The + strand is shown above the line, and the - strand is shown below. Genes are color-coded: RNA polymerase (blue), Ribosomal proteins (green), Other protein-coding (yellow), LSU rRNA (pink), SSU rRNA (light blue), and tRNA (black). The position in kb is indicated on the x-axis.

Figure 6. Apicoplast genome, circular but drawn linear. Genes on the + strand sit above the line, genes on the - strand below.

A comparison of mitochondrial genomes of Hepatozoon species. Ribbons link the *cox1*, *cox3*, and *cob* genes between neighboring genomes. The species listed are H. catesbianae NC_044466, H. clamatae MN245142, H. griseisciuri MK452388, H. griseisciuri MK452389, Hepatozoon sp. MN245143, and Hepatozoon sp. (this study). The position in kb is indicated on the x-axis.

Figure 7. Mitochondrial genomes of *Hepatozoon*. Ribbons link *cox1*, *cox3*, and *cob* between neighboring genomes.

Conclusions

- B. sazimai* carries a *Hepatozoon* lineage that differs from every compared species in gamont shape, 18S rRNA, and mitochondrial genome.
- Its mitochondrial genome is the largest reported for the genus, with a gene order not seen before.
- Two 18S variants in one host point to divergent rDNA copies or a mixed infection.
- Host genome projects also sequence blood parasites, at no extra cost.

Acknowledgments. We thank Instituto Butantan, Fundação Butantan, and Instituto de Biociências (USP) for institutional support, NBBC (Instituto Butantan) and UNC Charlotte for computing and sequencing, CAPES for funding (Finance Code 001), and the Sociedade Brasileira de Genética for organizing Genética 2026. Animal use: CEUAIB protocol 8447060524. Genetic heritage access: SISGEN A4A6CA4.

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